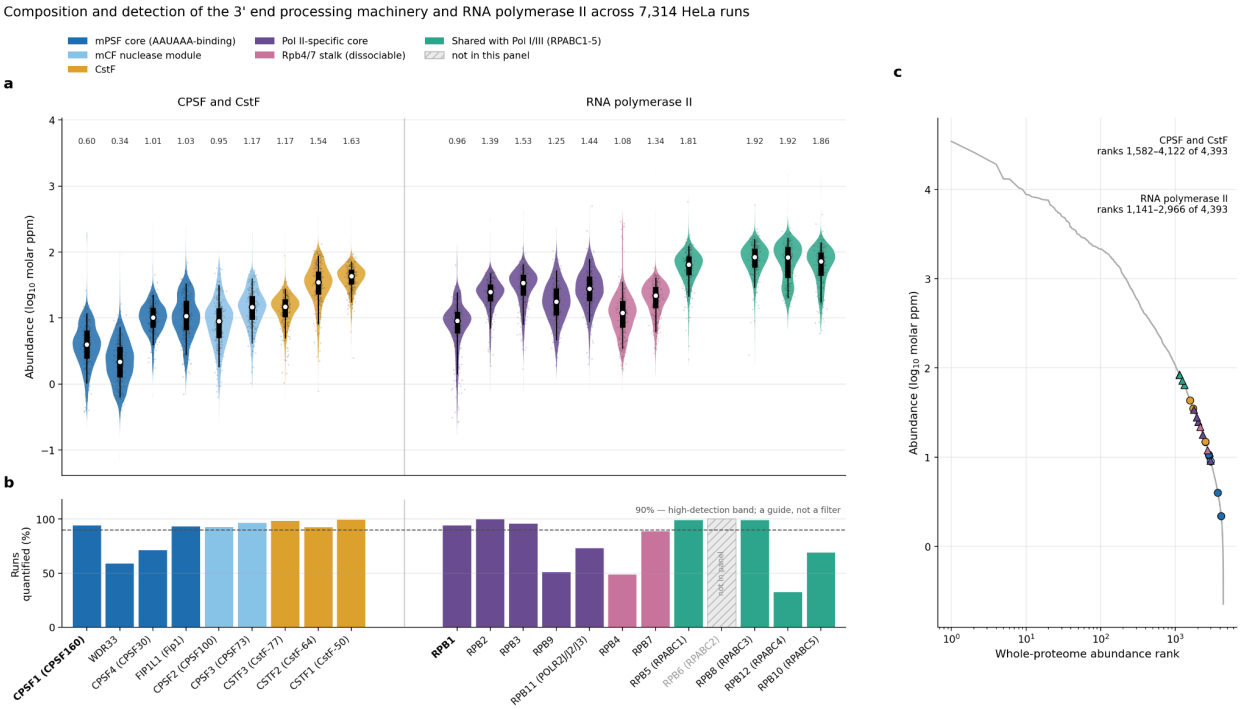
Results

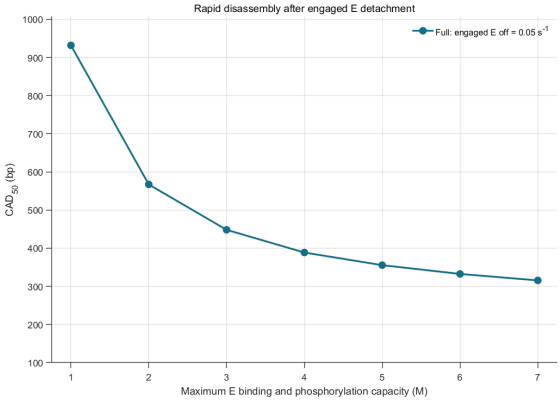
Genomics data show a comparable abundance of Polymerase II to CPA factors

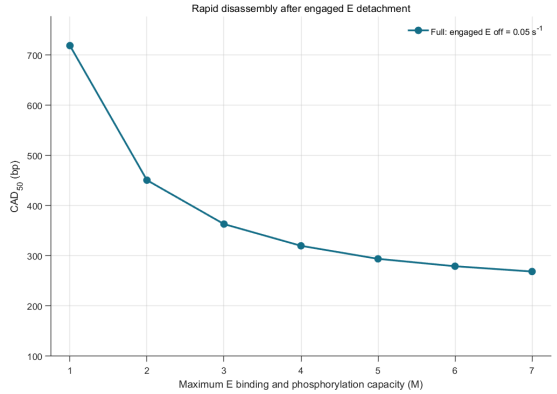


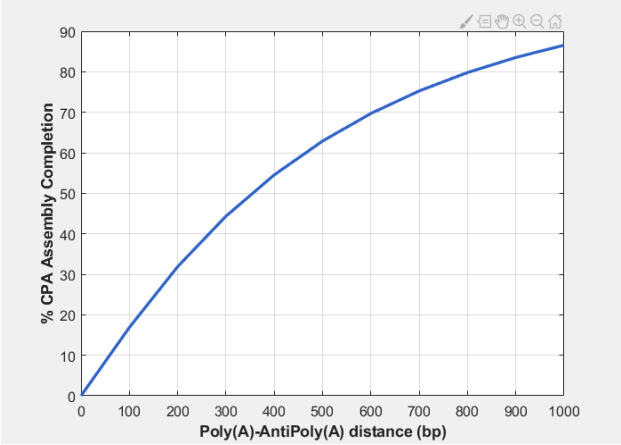
Multivalent CTD scaffolding reduces the distance required for CPA assembly

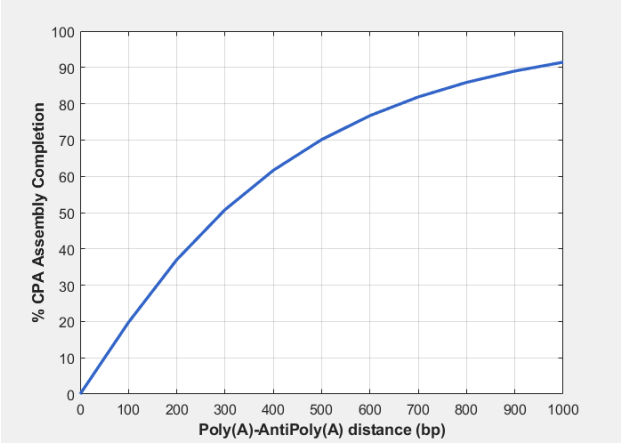
We first tested whether a single early-factor binding site on Pol II was sufficient to generate the short downstream assembly window expected for CPA. When each polymerase was restricted to a single E factor, the model produced a CAD of 1273 bp, which exceeded the expected 400-800 bp assembly window. Increasing the maximum number of E-binding sites progressively shortened CAD; a five-site scaffold reduced CAD to 436 bp and a six-site scaffold to 397 bp. Thus, multiple CTD-binding and preloading of the early CPA factor provide a mechanism for accelerating PAS commitment.

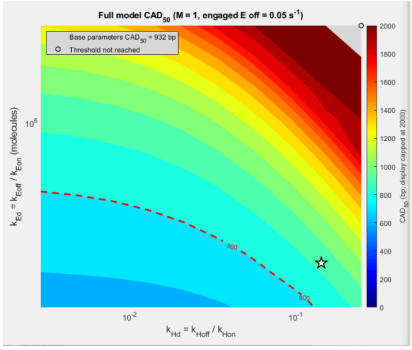
The occupancy profiles show why this effect emerges. Allowing more E-binding events increases the mean number of E factors carried by Pol II near the PAS, whereas the Ser2P profile remains defined by the upstream phosphorylation program. Parameter sweeps also indicated that the one-site regime remains inefficient over a broad range of binding parameters, supporting the interpretation that multivalent E-factor recruitment, rather than a single high-affinity interaction, is required for rapid CPA assembly.

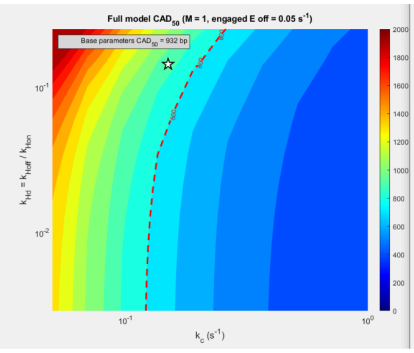

Rapid disassembly after engaged E detachment
Full: engaged E off = 0.05 s-1
CAD50 (bp)
1000
900
800
700
600
500
400
300
200
100
1
2
3
4
5
6
7
Maximum E binding and phosphorylation capacity (M)

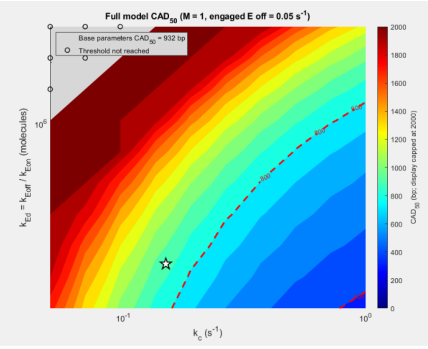

Rapid disassembly after engaged E detachment
Full: engaged E off = 0.05 s-1
CAD50 (bp)
700
600
500
400
300
200
100
1
2
3
4
5
6
7
Maximum E binding and phosphorylation capacity (M)

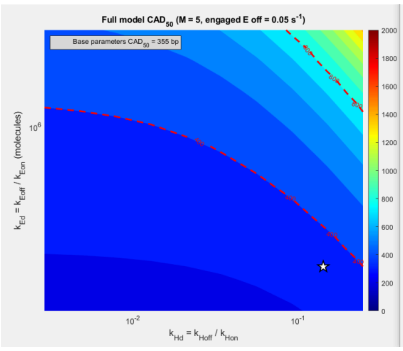

% CPA Assembly Completion
Poly(A)-AntiPoly(A) distance (bp)

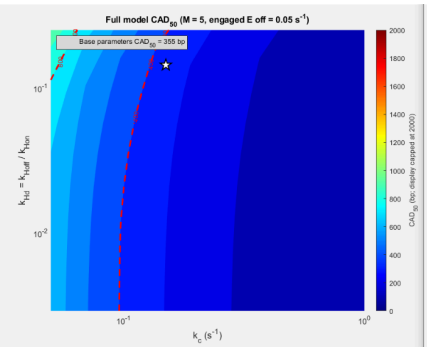

% CPA Assembly Completion
Poly(A)-AntiPoly(A) distance (bp)

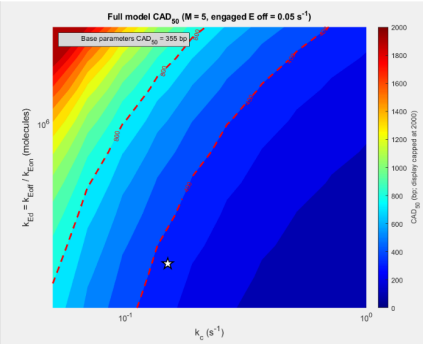

Full model CAD50 (M = 1, engaged E off = 0.05 s-1)
Base parameters CAD50 = 932 bp
Threshold not reached


Full model CAD50 (M = 1, engaged E off = 0.05 s-1)
Base parameters CAD50 = 932 bp


Full model CAD50 (M = 1, engaged E off = 0.05 s-1)
Base parameters CAD50 = 932 bp
Threshold not reached


Full model CAD50 (M = 5, engaged E off = 0.05 s-1)
Base parameters CAD50 = 355 bp

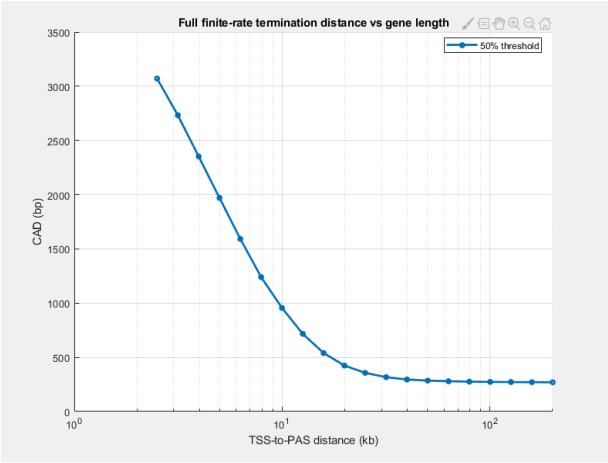

Full model CAD50 (M = 5, engaged E off = 0.05 s-1)
Base parameters CAD50 = 355 bp

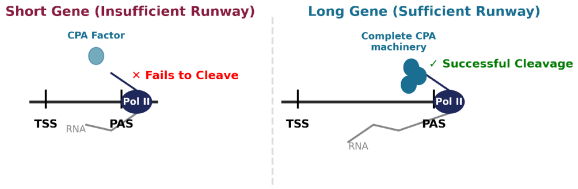

Full model CAD50 (M = 5, engaged E off = 0.05 s-1)
Base parameters CAD50 = 355 bp

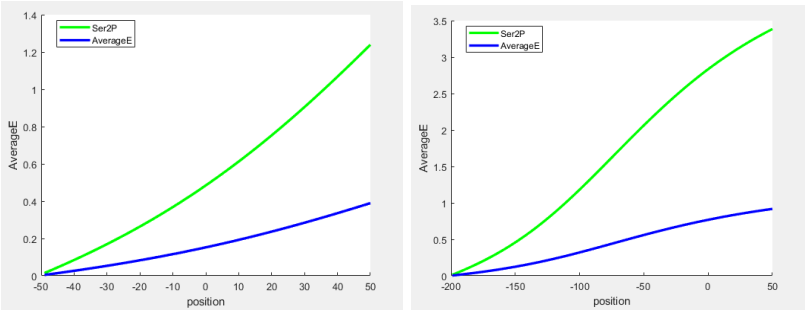
Gene length creates a pre-PAS loading runway for CPA factors

We next varied the distance between the TSS and the PAS to ask whether gene architecture affects assembly efficiency. CAD was highest for short TSS-to-PAS distances and decreased sharply as the upstream transcribed region increased. In the simulated parameter set, CAD approached a plateau near 400 bp once the gene length exceeded the human medium gene length (20k bp), indicating that the PAS was sufficiently far from the TSS.

These results suggest a pre-PAS loading-runway effect. On short genes, Pol II reaches the PAS before substantial Ser2P accumulation and E-factor loading have occurred, so commitment must be completed farther downstream. For longer genes, the polymerase arrives at the PAS with a more mature CTD binding CPA-factor state, allowing productive assembly over a shorter downstream distance.



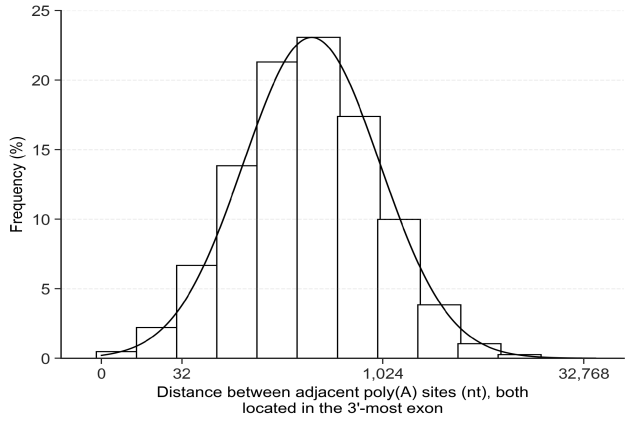




(merge the two into one)

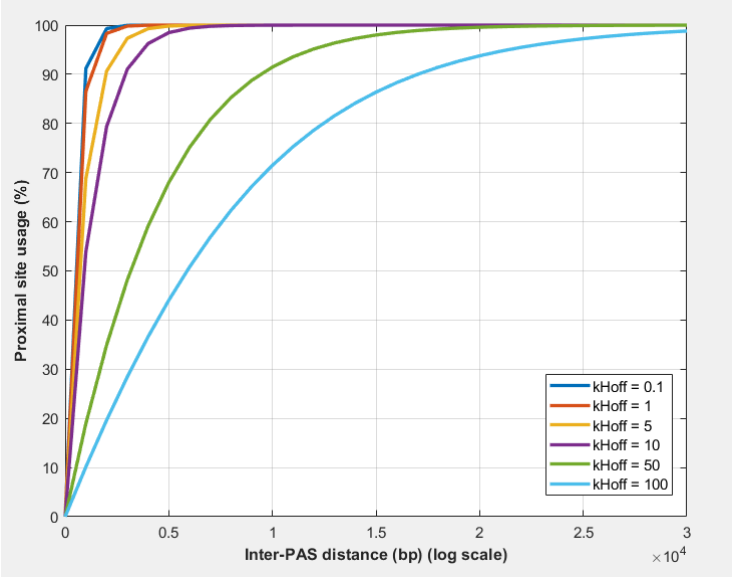
Termination CDFs recapitulate the logic of cis-antisense rescue assays

The cis-antisense rescue assay asks how far Pol II must travel after a PAS for the transcript to become resistant to downstream interference. In the model, this quantity maps directly to the termination CDF: the fraction committed by a given downstream distance is interpreted as the fraction protected from antisense disruption. The same CDF also provides a mechanistic estimate of proximal PAS usage in APA, because a polymerase will use the proximal PAS only if it commits before reaching the downstream PAS. For the most common inter-PAS distance in human genes (300 bp), proximal usage is intermediate, at around 40%, rather than fixed at 0% or 100%, making APA tunable to both PAS strength and factor availability.



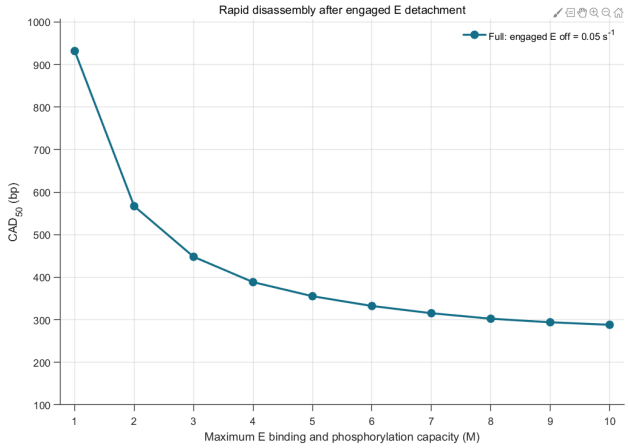
APA tunability persists across a wide PAS-strength range

We then asked whether this tunability persisted when the inter-PAS distance and PAS strength was varied. Sweeping the PAS dissociation rate shifted the proximal-usage curves over a broad range: stronger PASs reached high proximal usage over short inter-PAS distances, whereas weaker PASs required longer spacing and often remained only partially used. The qualitative dependence on distance was retained across the sweep, indicating that PAS spacing and PAS strength jointly determine the APA response.



Resource competition

Increasing the number of E binding



- Increasing the number of E binding increases the global competition, but also increases local assembly efficiency (local efficiency outweighs global competition in this case). This combined effect results in a plateau of CAD when an increasing number of E binding.

Global competition is affected by the E_total and Pol_total

